

The Effect of COVID-19 on the Well-Being of First-Year University Students

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Abstract

Past studies have found that university students' well-being is relatively low, and that first-year students are specifically vulnerable. This issue has drawn particular attention during the COVID-19 pandemic. This study examines the effect of COVID-19 on the well-being, academic confidence, feeling of informedness, and self-efficacy of first-year university students. Both cross-sectional and longitudinal data were used. Survey data were collected from two cohorts of first-year students at a Belgian university ($N = 997$): a pre-covid cohort ($N = 493$) and a covid-affected cohort ($N = 504$). Data was gathered at two measurement moments for the pre-covid cohort and at four measurement moments for the covid-affected cohort. First, between-subjects analyses (ANCOVA) revealed no significant effect of COVID-19 on students' well-being ($p > .100$), academic confidence ($p > .100$), and feeling of informedness ($p > .100$), on arrival at university. Three weeks later, however, the covid-affected cohort reported (marginally) lower well-being than the pre-covid cohort ($p = .057$, $\eta^2 = .01$). Second, longitudinal analyses (repeated measures ANCOVA) on the covid-affected cohort revealed changes throughout the academic year in well-being ($p < .001$, $\eta_p^2 = .39$) and self-efficacy ($p < .001$, $\eta_p^2 = .04$). Well-being and self-efficacy decreased during the first semester ($p < .001$), and slightly recovered at the start of the second semester ($p < .001$). Educational implications on how to support students during their transition to higher education are discussed.

Keywords: COVID-19; first-year students; higher education; well-being; self-efficacy

Word count: 8,555 words

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Numerous studies show that university students suffer from low well-being (e.g., Cooke et al., 2006; Larcombe et al., 2016). In line with this, scholars agree on the alarming prevalence of mental health issues among higher education students: roughly one out of four suffers from at least one mental health problem (Auerbach et al., 2018; Larcombe et al., 2016). Hence, student well-being has become a pressing concern in higher education contexts, as universities are uniquely positioned to support their students' well-being (Baik, et al., 2019). First-year undergraduate students are particularly susceptible to mental distress (Auerbach et al., 2018). Their arrival at university is often accompanied by challenges to adjust and integrate, such as leaving home, making new friends, handling academic expectations, and developing new learning styles (Conley et al., 2014; Vinson et al., 2010). Struggling with academic, social, and personal-emotional adaptation is closely related to students' well-being (St Clair-Thompson et al., 2017), which, in turn, affects learning experiences and academic satisfaction (Baik et al., 2019), academic performance (Parada & Verlhac, 2022) and university degree completion (Holliman et al., 2018).

Aside from well-being, psychological literature has identified other concepts essential to a smooth university transition. This study focuses on first-year students' self-efficacy, academic confidence, and feeling of informedness. *Self-efficacy* pertains to an individual's overall belief in one's own capabilities to perform tasks (i.e., self-reliance), to cope with difficulties (i.e., resilience), and to control the way in which one acts (i.e., self-control) (Bandura, 1977, 1997), and is therefore a personal resource associated with well-being (Gutiérrez & Tomás, 2019; Lau et al., 2019). Greater self-efficacy facilitates university adjustment, with students perceiving their transition as challenging rather than threatening or stress inducing (Morton et al., 2014; Sánchez-

Teruel et al., 2021). *Academic confidence* refers to students' beliefs in their own abilities to succeed academically, and thus relates specifically to one's self-estimated performance at university (Stankov et al., 2014). In this way, academic confidence is connected to well-being during students' transition, and therefore to their personal-emotional adjustment to university (Michie et al., 2001; St Clair-Thompson et al., 2017). Students' *feeling of informedness* involves the degree to which students feel sufficiently informed on and prepared for their university transition, by, for example, having the necessary information on their study program (Mouton & Ertl, 2021). This feeling of informedness during university transition relates to favorable outcomes, such as higher GPA, higher learning satisfaction, and retention (Lowe & Cook, 2003; Mouton & Ertl, 2021).

Considering these academic findings, it should be a top priority for universities to facilitate first-year students' transition to higher education, and to promote their well-being. Therefore, the university in this study decided to collect data on students' well-being during their first weeks of university (two measurement moments).

Notwithstanding, the urgency of monitoring students during their university transition has intensified since the COVID-19 outbreak, as this has put further pressure on student well-being, especially of first-year students (Dodd et al., 2021). Therefore, at the university in this study, data collection on student well-being during the pandemic was extended to four measurement moments throughout the academic year.

The urgency to monitor students' well-being in times of COVID-19 is highlighted by several studies, which focused on the first home confinement period in spring 2020 (e.g., Husky et al., 2020). According to Dodd et al. (2021), only 35% of students reported a "sufficient level of well-being" during the first months of the pandemic, and 32% even reported "very low well-being". This high susceptibility to

psychological distress was due to profound changes in students' social life and living and working conditions. Other pandemic-induced stressors associated with mental problems were uncertainty about academic progress, increased academic workload, financial concerns, and loneliness (Tasso et al., 2021). Importantly, COVID-19 has placed even greater strain on the well-being of first-year students (Dodd et al., 2021), the target group of the present study, hampering their already stressful transition and adding even more uncertainty to their process of university adjustment (Kwan et al., 2021). For students who completed secondary education during the pandemic, preparatory initiatives (e.g., information fairs) and orientation activities (e.g., welcoming days) were canceled during their transition period, or only organized online (Pownall et al., 2022). Later, after arrival at university, on-campus activities were hardly allowed during these students' first weeks and months, due to pandemic-related restrictions. Moreover, the disruptive switch to e-learning and subsequent social isolation elicited feelings of frustration, demotivation, alienation, and emotional exhaustion (Juntunen et al., 2022; Laher et al., 2021). All of this translated into few opportunities for students to assimilate into the educational climate and to make new friends—dynamics that potentially affect well-being, self-efficacy, academic confidence and feeling of informedness.

This study contributes to the literature on student well-being in several ways. First, it examines the effects of the pandemic in the fall semester of 2020, several months after the initial COVID-19 outbreak (March 2020). As Charles et al. (2021) indicated, many scholars have focused only on the instantaneous effect of the first lockdown in spring 2020 on student well-being (e.g., Savage et al., 2020), rather than the second COVID-19 wave. This study includes students who arrived at university without having finished secondary education in a traditional, pre-pandemic manner.

Second, this study focuses on first-year students, a subgroup particularly susceptible to mental distress (Auerbach et al., 2018). Third, this study aims to follow the recommendations of Husky et al. (2020) and Sánchez-Teruel et al. (2021), to conduct longitudinal research on student well-being in times of COVID-19, and combines this with a cross-sectional design. The cross-sectional design includes between-subjects analyses and compares a covid-affected student cohort with a pre-covid cohort, during the transition period at the beginning of an academic year (2×2 measurement moments). The longitudinal within-subjects design offers the opportunity to analyze the well-being and self-efficacy of a covid-affected cohort across several months, by means of four measurement moments. Fourth, this study covers a Western European (Belgian) university context, and thereby focuses on an area which has remained understudied in higher education literature on student well-being. Universities in Belgium have an open admission system, implying greater heterogeneity among first-year students than many other studies. Finally, the added value of this study is not limited to the specific educational implications of the pandemic. Whilst promoting student well-being was particularly urgent during the COVID-19 crisis, this topic will—and should—remain relevant in the future. Consequently, this study highlights the need for higher education institutions to rethink their policies on how to support the well-being and the university transition of first-year students.

The Effect of COVID-19 on the Transition to University

Part of this study's sample comprises covid-affected students, who were in their final semester of secondary education during the first COVID-19 outbreak. Consequently, these students finished secondary education in times of e-learning, and their preparation for higher education was hampered, leading to increased academic pressure and impaired *well-being* during their university transition (Giannopoulou et al.,

2021; Kwan et al., 2021). Once these first-year students started university, the establishment of social connections—which can be facilitated by physical orientation activities—was hampered by pandemic-related measures, resulting in increased anxiousness and loneliness (Dodd et al., 2021; Juntunen et al., 2022). Hence, the university transition of these students was severely disrupted, both by the long-term negative effect of the pandemic during secondary education, and by the impact of the public health measures at the beginning of their university journey (Pownall et al., 2022). Giving these dynamics, H1a expects a difference in well-being between the covid-affected and the pre-covid cohort.

H1a: On arrival at university, the well-being of the covid-affected student cohort is lower, compared to the pre-covid student cohort.

The next hypothesis (H1b) involves *academic confidence*, which comprises students' beliefs in their ability to succeed academically, and which is closely related to their well-being and learning success during university transition (Michie et al., 2001; St Clair-Thompson et al., 2017). Furthermore, first-year students' drop-out depends on their academic confidence when starting university (Mouton et al., 2020). Because of the pandemic, students were offered fewer opportunities to prepare for their transition: final exams in secondary school and information days regarding higher education were canceled. According to Pownall et al. (2022), this lack of preparation activities threatened the academic confidence of first-year students.

H1b: On arrival at university, the academic confidence of the covid-affected student cohort is lower, compared to the pre-covid student cohort.

The next hypothesis (H1c) involves first-year students' *feeling of informedness*, a concept closely related to learning outcomes and academic success of first-year students (Lowe & Cook, 2003). While preparing their university transition, many

students risk increased academic pressure, insecurity and stress, which may compromise their feeling of informedness (Giannopoulou et al., 2021). Furthermore, the insecurity and information deficits induced by COVID-19, were fundamental antecedents of students' psychological distress (Sánchez-Teruel et al., 2021; Wen et al., 2021).

Additionally, as mentioned above, first-year students had fewer opportunities to obtain preparatory information, since information fairs and other pre-registration events were canceled or only online. Pownall et al. (2022) argued that students thus possessed less knowledge about university and its unspoken rules. Therefore, it is expected that covid-affected first-year students felt less prepared during their university transition.

H1c: On arrival at university, the feeling of informedness of the covid-affected student cohort is lower, compared to the pre-covid student cohort.

Transition to higher education has often been examined in terms of students' well-being during the first few weeks or months at university (e.g., Conley et al., 2014; Cooke et al., 2006). This transition cannot be considered as an isolated induction moment on arrival, but as a diffusive process during the initial weeks and months, including social integration and various adjustment strategies (Brooman & Darwent, 2014). Previous literature identified several underlying mechanisms of settling down, socially integrating, and gaining confidence, during the first few weeks at university (Vinson et al., 2010). Consequently, H1d includes the well-being of transitioning students beyond the moment of arrival at university. Analogous to H1a, a lower well-being is expected for the covid-affected cohort, three weeks after arrival at university.

H1d: Three weeks after arrival at university, the well-being of the covid-affected student cohort is lower, compared to the pre-covid student cohort.

The Longitudinal Effects of COVID-19 on the Well-Being and Self-Efficacy of First-Year Students

Universities have been recommended to monitor students' well-being longitudinally in times of COVID-19 (Husky et al., 2020; Sánchez-Teruel et al., 2021). Research indicated a harmful 'duration effect' of governmental COVID-19 restrictions: the longer public health measures have been imposed, the greater the negative mental repercussions (e.g., stress, anxiousness) on students (de Man et al., 2021; Husky et al., 2020; Savage et al., 2020). This negative effect on student well-being has been documented during the first COVID-19 outbreak in spring 2020 (e.g., de Man et al., 2021). However, this study investigates the second wave of COVID-19 contaminations, during fall and winter 2020. After summer, pressure on health care systems again increased from September 2020 onwards, and governmental policies gradually reintroduced more stringent restrictions for higher education institutions (e.g., social contact constraints, limited on-campus learning opportunities, physical distancing). Consequently, public health regulations against the spread of COVID-19 could have affected how student well-being changed during the pandemic's second wave, as stated in H2a.

H2a: The well-being of students decreases throughout a covid-affected academic year.

Finally, H2b involves *self-efficacy*, a trait which may protect first-year students from mental problems induced by challenging educational circumstances (Lau et al., 2019), and which facilitates their adjustment to the university context (Morton et al., 2014). Next to its fundamental association with well-being, self-efficacy also contributes to students' academic success (Gutiérrez & Tomás, 2019). In light of COVID-19, students' resource of self-efficacy proved especially valuable: highly self-efficacious students were more proficient at coping with the adversities of the

pandemic, as self-efficacy mitigated their stress and contributed to improved well-being (e.g., Sánchez-Teruel et al., 2021; Wen et al., 2021). Nonetheless, students' self-efficacy was directly weakened by the pandemic (Alemany-Arrebola et al., 2020). Given the assumed longitudinal effect of COVID-19 on well-being, a similar trend is expected for self-efficacy.

H2b: The self-efficacy of students decreases throughout a covid-affected academic year.

With answers to these hypotheses, the present study aims to inform educational literature and educational practice, by creating knowledge to support the well-being of first-year students, while they face the challenges that come with their transition to higher education. This concern has been highlighted since COVID-19, but certainly remains relevant after the pandemic.

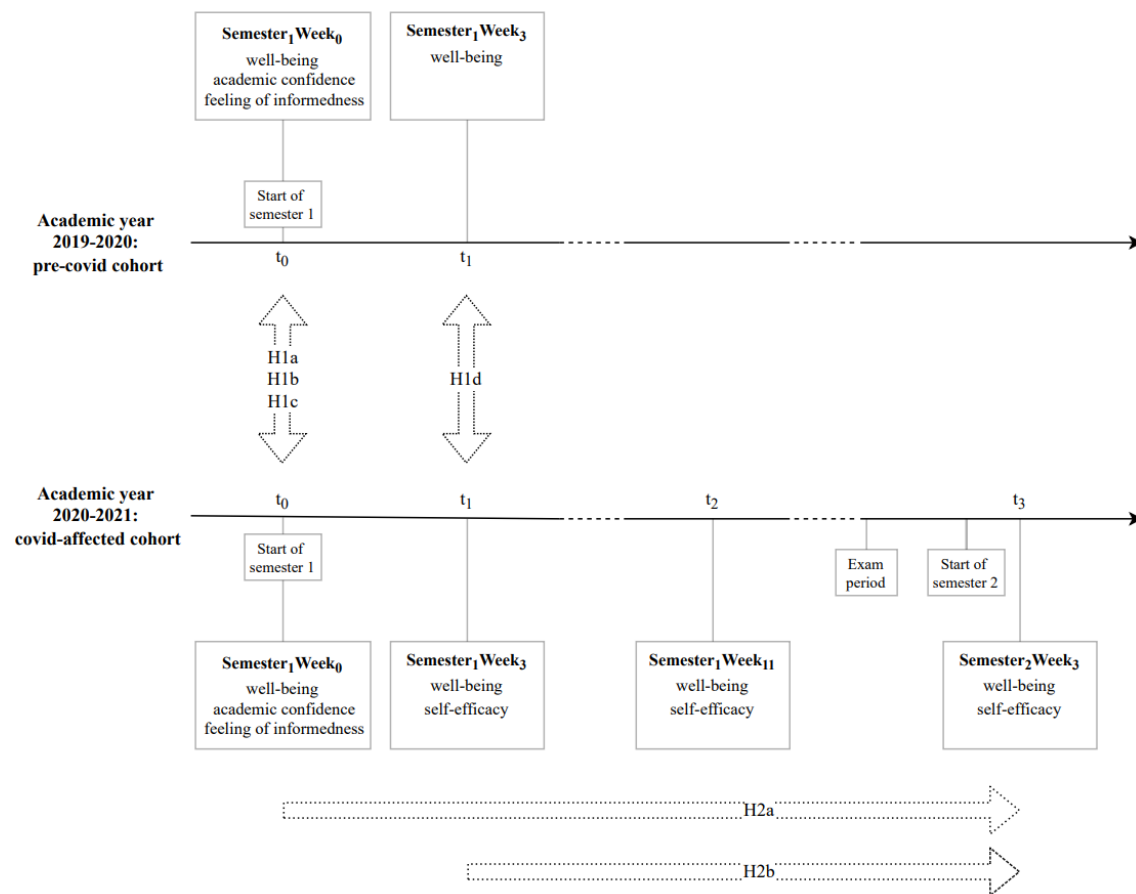
Methodology

Study Design

The study design covered a pre-covid period and a covid-affected student sample. The pre-covid sample, referred to as the *pre-covid student cohort*, participated before the pandemic. The second sample involved a *covid-affected student cohort*: students who enrolled at university in September 2020. As the COVID-19 outbreak in Belgium occurred in March 2020, the covid-affected cohort was in their final semester of secondary education during the first national lockdown. Consequently, the covid-affected cohort had a history of coping with pandemic-related restrictions before they arrived at university. Moreover, throughout the period this covid-affected cohort was studied, governmental and university restrictions changed (see Appendix A for

education-specific restrictions). These nation-specific pandemic-related restrictions were similar to other European countries (Marinoni et al., 2020).

This study comprises an incomplete split-plot 2×4 factorial design (Appendix B), including a cross-sectional 2×2 factorial between-subjects design (H1a–H1d) and a longitudinal 1×4 within-subjects design (H2a–H2b), in a field study context. Figure 1 illustrates the research design, hypotheses, measurement moments, and variables. First, quantitative data were collected from the pre-covid cohort—in light of a prior deliberate decision to monitor students' well-being and their university transition—at two measurement moments: the week before the start of the first semester (Semester₁Week₀ in September 2019, or t_0) and the third week of the first semester (Semester₁Week₃ in October 2019, or t_1). These measurement moments serve to test the first hypotheses (H1a–H1d). Second, the covid-affected cohort was studied at four measurement moments: aside from t_0 (Semester₁Week₀ in September 2020) and t_1 (Semester₁Week₃ in October 2020), additional data collection moments were organized in the eleventh week of the first semester (Semester₁Week₁₁ in December 2020, or t_2) and the third week of the second semester (Semester₂Week₃ in February 2021, or t_3). The four measurement moments during the covid-affected period contribute to testing the second hypotheses (H2a–H2b).

Figure 1*Research design*

An academic year at the university in question consists of two semesters of 12 weeks. The first semester (fall) starts in the third week of September. In December, students have a three-week Christmas break to study for the January exams. The second semester (spring) starts after spring break in February and lasts until May. The second exam period is in June. The resit examination period is in August. The choice for the measurement moments (t_0 through t_3) were inspired by the work of Vinson et al. (2010) and Cooke et al. (2006), who offer insights on student well-being at the start and end of the first semester, and at the beginning of the second semester. The measurement moments also correspond to different pandemic-related restrictions (Appendix A).

Participants

This study was conducted during academic years 2019–2020 and 2020–2021 in a first-year bachelor's program at the Faculty of Economics and Business Administration of a large Belgian university. Since this program is taught in Dutch (the university's official language), no international students participated in this study. Belgian higher education institutions follow an open-access system, with no formal admission procedures. Tuition fees are relatively low (approximately US\$1,150 per year), since Belgian universities is largely publicly financed. Therefore, first-year students constitute a heterogeneous group in terms of their secondary education background and socio-economic status.

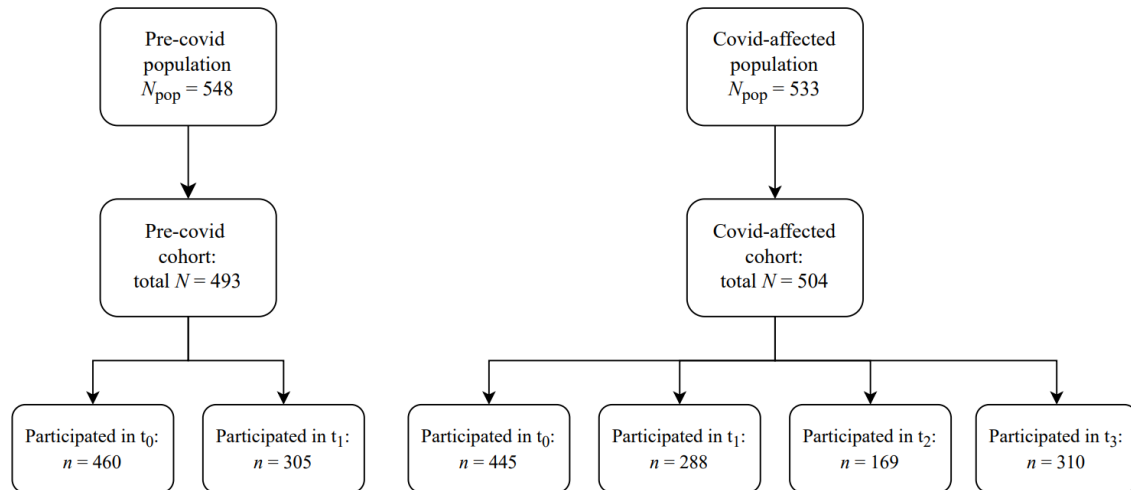
Figure 2 shows how many students participated in the separate data collection moments. The total sample comprises 997 unique participants (92% of the total population comprising the first-year program in the pre-covid and covid-affected periods): 192 female and 299 male students in the pre-covid cohort, 199 female and 302 male students in the covid-affected cohort, and five students not sharing their gender. Both student samples are highly comparable in terms of gender distribution ($\chi^2(1, 992) = 0.04, p = .842$), residency status (dormitory or commuting) ($\chi^2(1, 893) = 0.98, p = .322$), and secondary education background in mathematics ($t(901) = 0.51, p = .609$), economics ($t(902) = -1.25, p = .211$), and accounting ($t(899) = 0.06, p = .952$). These sociodemographic similarities contribute to the internal validity of the cross-sectional comparisons between student cohorts.

A substantial number of students participated in more than one measurement moment. In the pre-covid period, 272 students (50% of the population) completed both surveys. In the covid-affected period, 381 students (71% of the population) completed

at least two surveys; 98 (18% of the population) completed all four surveys. Appendix C illustrates the attrition within both cohorts.

Figure 2

Sample breakdown



Note. t₀ = Semester₁Week₀, t₁ = Semester₁Week₃, t₂ = Semester₁Week₁₁, t₃ = Semester₂Week₃.

Measurement Instruments

Every data point is characterized by the cohort (pre-covid or covid-affected), the measurement moment (t₀, t₁, t₂, or t₃), and the student's gender. Figure 1 shows the timeline of the data collected at each measurement moment. Appendix D presents the survey items. One of the central variables is well-being, which, drawing on Stankov et al. (2014), was measured by using six items of the Warwick-Edinburgh Mental Well-being Scale (WEMWBS), developed by Tennant et al. (2007). Second, Stankov et al.'s (2014) single-item measure of 'confidence in academic achievement' served as the measure of academic confidence in this study. Students were asked to estimate their own likelihood of succeeding in the first undergraduate year. Furthermore, Lörz et al.'s (2012, cited in Mouton & Ertl, 2021) single-item measure of feeling 'overall well prepared for university studies' was adapted to measure students' feeling of

informedness. As such, the survey at t_0 included two single-item measures, entirely based on previous literature. Finally, self-efficacy was assessed with the General Self-Efficacy Scale (GSES) of Schwarzer and Jerusalem (1995) and was examined only for the covid-affected cohort. Both the WEMWBS and GSES have been validated in similar research contexts of university students in times of COVID-19, respectively by, among others, Savage et al. (2020) and Sánchez-Teruel et al. (2021). Cronbach alpha coefficients for both scales suggest good internal consistency (Table 1).

Procedure for Data Collection

Online surveys were conducted at all data collection moments. The pre-covid cohort was invited to participate in two surveys (t_0 and t_1), the covid-affected cohort in four surveys (t_0 through t_3). Figure 1 presents these timelines. In both periods, the first survey was administered during an orientation activity at the start of the academic year (t_0). The other surveys (t_1 through t_3) were introduced during class time. Participation was always voluntary. Data for different surveys within the same cohort were combined using student IDs. All respondents gave their active informed consent. Students did not enter their names, were assured confidentiality, and were guaranteed that the data would be used for research purposes only. The data of different surveys were linked by an external researcher using student IDs, and the initial data were stored separately from the anonymous research files.

Procedure for Data Analysis

Several steps were involved in the data analysis. First, descriptive statistics and correlations are presented. Subsequently, analyses of covariance (ANCOVA) are provided to test each hypothesis. To test H1a–H1d, within-subjects ANCOVA are performed to compare the pre-covid cohort and the covid-affected cohort, resulting in

estimated marginal means. Moreover, to test H2a–H2b, repeated measures ANCOVA (between-subjects) serve to analyze the changes in well-being and self-efficacy of the covid-affected cohort.

Results

Descriptive Statistics

Table 1 presents descriptive statistics. Well-being at t_0 was, on average, higher than at t_1 for both cohorts. The differences between student cohorts are rather small. It can also be seen that the covid-affected cohort experienced decreasing well-being during the first semester (t_0 through t_2), followed by an increase in the second semester (t_2 to t_3). Self-efficacy followed the same fluctuations, although this is less apparent from Table 1.

Table 1*Descriptive statistics, internal consistency, and correlations*

Variable	Pre-covid			Covid-affected			Total sample correlations									
	M	SD	a	M	SD	a	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1. WB t ₀	4.06	0.57	.79	4.10	0.54	.76	1.00	.29**	.30**	.56**	.33**	.36**	.42**	.42**	.39**	-.14**
2. AC t ₀	3.40	0.73	-	3.47	0.68	-		1.00	.16**	.22**	.17	.04	.25**	.20*	.13*	-.14**
3. FoI t ₀	3.88	0.91	-	3.81	0.85	-			1.00	.21**	.08	.07	.10	-.03	.01	.00
4. WB t ₁	3.71	0.58	.72	3.66	0.56	.74				1.00	.44**	.41**	.42**	.33**	.31**	-.18**
5. WB t ₂				3.02	0.67	.72					1.00	.46**	.19**	.44**	.35**	-.19*
6. WB t ₃				3.57	0.76	.80						1.00	.29**	.38**	.35**	-.11
7. SE t ₁				3.13	0.33	.77							1.00	.54**	.61**	-.22**
8. SE t ₂				3.03	0.42	.84								1.00	.67**	-.17*
9. SE t ₃				3.12	0.42	.84									1.00	-.17**
10. Gender				-	-	-										1.00

Note. WB = well-being, AC = academic confidence, FoI = feeling of informedness, SE = self-efficacy.

t₀ = Semester₁Week₀, t₁ = Semester₁Week₃, t₂ = Semester₁Week₁₁, t₃ = Semester₂Week₃. Gender is coded as 0 = male, 1 = female.

a = Cronbach alphas, which are in line with previous literature (e.g., Lau et al., 2019).

* $p < .05$. ** $p < .01$

Table 1 also shows Pearson correlation coefficients. Unsurprisingly, measures from same measurement moment are highly correlated. At t_0 , well-being is positively correlated with academic confidence and feeling of informedness. Well-being and self-efficacy are systematically positively correlated at the same measurement moments. Well-being measures are significantly positively intercorrelated; the same applies for self-efficacy. Academic confidence at t_0 positively correlates with the self-efficacy measures. However, the correlations between these variables are not excessively high, which underlines the theoretical distinction between both concepts. Finally, the correlations indicate gender differences. Female students generally have lower well-being, academic confidence, and self-efficacy. Gender is included as control variable when testing the hypotheses.

Hypotheses Testing

To test H1a–H1c, the pre-covid and covid-affected cohorts were compared at the start of the academic year (t_0). ANCOVA, presented in Table 2, did not demonstrate significant differences regarding well-being, academic confidence, and feeling of informedness, while controlling for gender. The data indicated that students who experienced a pandemic-affected semester in secondary school, arrived at university with the same mental resources as pre-covid students. Therefore, H1a (well-being), H1b (academic confidence), and H1c (feeling of informedness) are not supported by the data.

Hypothesis 1d presumed a difference in well-being at t_1 between the pre-covid and covid-affected cohort. Gender and well-being at t_0 were included as control variables (Table 2). The latter control variable (initial level of well-being at t_0) was included to cover a potential ceiling effect. The ANCOVA revealed that the covid-affected students reported marginally lower well-being at t_1 , compared to pre-covid

students. Hence, H1d is marginally supported. Additionally, paired samples *t*-tests indicated a significant decline in well-being from t_0 to t_1 for both cohorts (not tabulated: $t = 9.76, p < .001$ for the pre-covid cohort, $t = 12.79, p < .001$ for the covid-affected cohort).

Table 2

Estimated Marginal Means of Dependent Variables at t_0 and t_1 (H1a–H1d)

	Pre-covid	Covid-affected	<i>F</i> ratio	<i>p</i>	η^2
Well-being t_0 ^a	4.05	4.10	1.64	.201	.00
Academic confidence t_0 ^b	3.40	3.47	1.64	.200	.00
Feeling of informedness t_0 ^c	3.88	3.81	1.76	.185	.00
Well-being t_1 ^d	3.74	3.66	3.64	.057	.01

Note. t_0 = Semester₁Week₀, t_1 = Semester₁Week₃.

All analyses controlled for students' gender; the analysis on well-being t_1 (H1d) also controlled for well-being t_0 .

^a $n = 884$; Model statistics: $F = 9.37, p < .001, \eta_p^2 = .02, \text{Adj. } R^2 = .02$

^b $n = 782$; Model statistics: $F = 8.38, p < .001, \eta_p^2 = .02, \text{Adj. } R^2 = .02$

^c $n = 882$; Model statistics: $F = 0.88, p = .415, \eta_p^2 = .00, \text{Adj. } R^2 = .00$

^d $n = 520$; Model statistics: $F = 85.41, p < .001, \eta_p^2 = .33, \text{Adj. } R^2 = .33$

To address H2a–H2b, within-subjects analyses on the longitudinal dataset of covid-affected students are presented: repeated measures ANCOVA were performed on the entire covid-affected sample. However, a missing-data issue was encountered. 381 covid-affected students reported data on at least two measurement moments, yet only 98 participated in all four. For the subsample of 381 students, missing data were handled by the expectation-maximization algorithm for Maximum Likelihood Estimation (Schafer & Graham, 2002). Little's (1988) Missing Completely At Random (MCAR) test was used to compare participants with and without complete data ($\chi^2(190) = 202.08$,

$p = .261$). As the missing data were MCAR, a reliable imputation of missing values was allowed for (iteratively estimated from all non-missing values on all variables included in this study). Hence, the second step of longitudinal analyses (H2a–H2b) involved repeated measures ANCOVA after missing value estimation using the expectation-maximization algorithm.

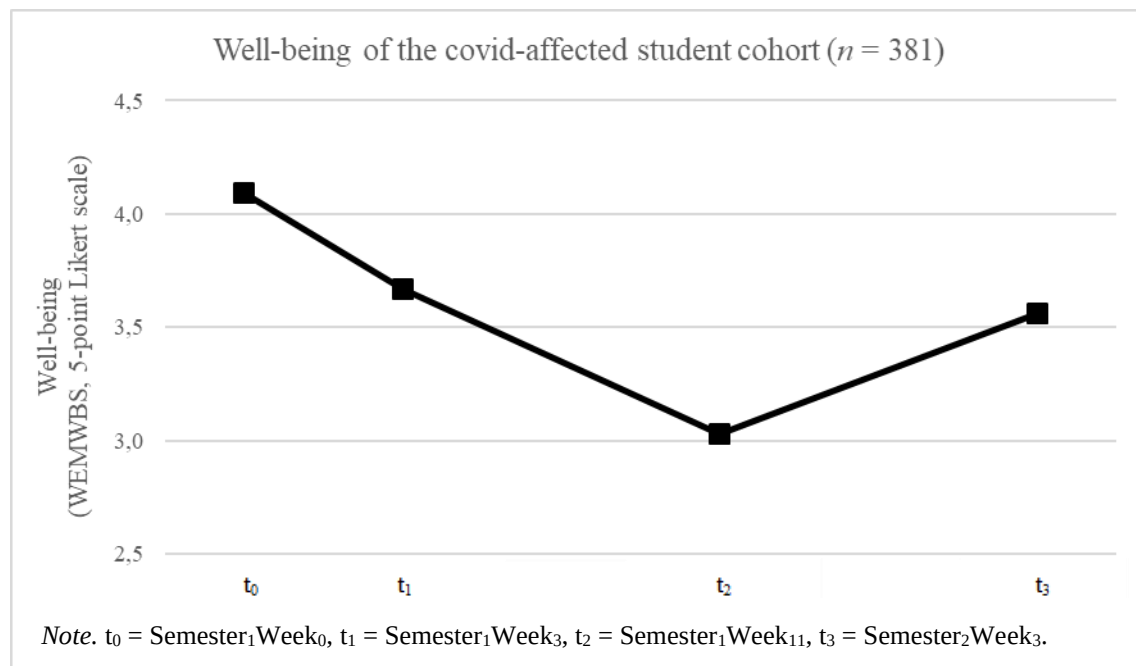
Regarding H2a, a repeated measures ANCOVA showed that the well-being of covid-affected students changed significantly over time ($F(2.50, 948.86) = 246.57, p < .001, \eta_p^2 = .39$), while controlling for gender. Bonferroni pairwise comparisons (Table 3, Panel A) indicated that well-being declined throughout the first semester (t_0 through t_2) and increased again from t_2 to t_3 . The latter dynamic implies that students recovered after the exam period. Figure 3 illustrates these estimated marginal (EM) means. In sum, H2a is partly supported, for the first semester only. Contrastingly, well-being did not decline further after the first semester; a resurfacing effect is indicated at the start of the second semester.

Table 3*Well-being and self-efficacy of the covid-affected student cohort (H2a–H2b)*

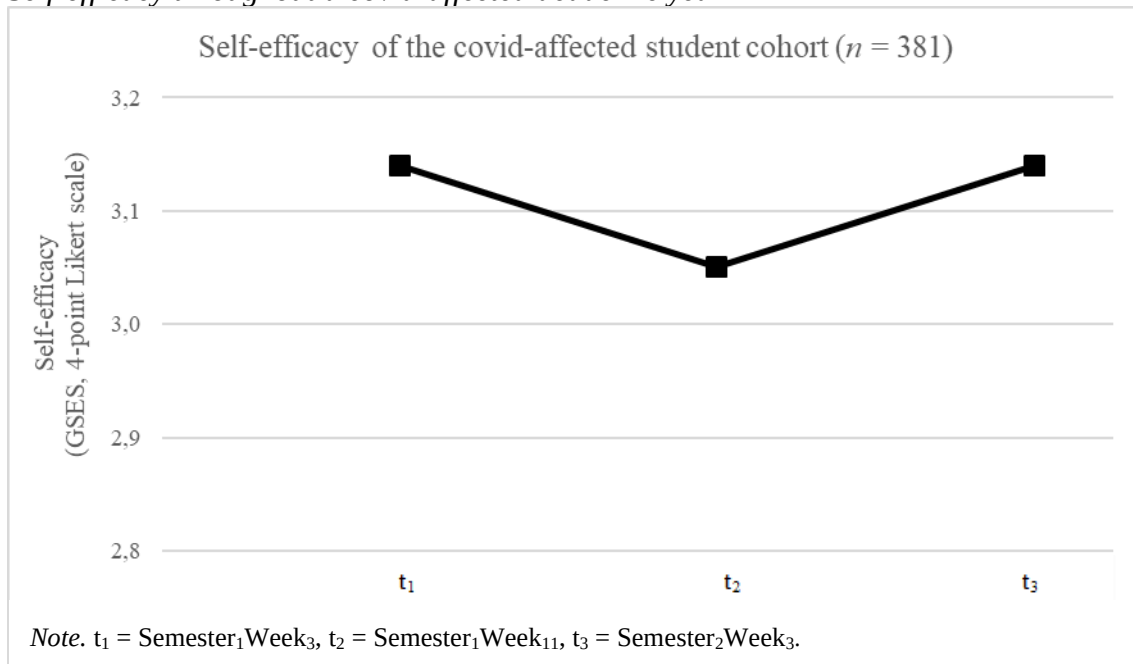
Panel A: Repeated-measures ANCOVA on well-being ($n = 381$)				
Time sample	EM Mean	Bonferroni post-hoc comparisons (p)		
		t_0	t_1	t_2
t_0	4.09			
t_1	3.67	< .001		
t_2	3.03	< .001	< .001	
t_3	3.56	< .001	.004	< .001

Panel B: Repeated-measures ANCOVA on self-efficacy ($n = 381$)				
Time sample	EM Mean	Bonferroni post-hoc comparisons (p)		
		t_1	t_2	
t_1	3.14			
t_2	3.05	< .001		
t_3	3.14	1.000		< .001

Note. t_0 = Semester₁Week₀, t_1 = Semester₁Week₃, t_2 = Semester₁Week₁₁, t_3 = Semester₂Week₃.

Figure 3*Well-being throughout a covid-affected academic year*

Regarding students' self-efficacy during the covid-affected period (H2b), analogous data analysis was applied. A repeated measures ANCOVA demonstrated a significant change in self-efficacy over time ($F(1.96, 742.03) = 15.34, p < .001, \eta_p^2 = .04$), while controlling for gender. Bonferroni pairwise comparisons (Table 3, Panel B) indicated a significant decline in self-efficacy from t_1 to t_2 , and a significant increase from t_2 to t_3 , to a level comparable to t_1 . Figure 4 illustrates the EM means. Hence, H2b is supported with respect to the first semester only: self-efficacy decreased from t_1 to t_2 . Moreover, similar to well-being, the data illustrated that self-efficacy increased after the first semester, from t_2 to t_3 .

Figure 4*Self-efficacy throughout a covid-affected academic year*

Discussion

Main Findings

This study focused on the effect of COVID-19 on first-year students' university transition, in terms of well-being, academic confidence, feeling of informedness, and self-efficacy. This study combined a cross-sectional and longitudinal design. The results can be summarized in two main findings. First, cross-sectional analyses did not reveal any negative repercussions of the pandemic on students' well-being, academic confidence, or feeling of informedness on arrival at university (H1a–H1c). Three weeks later, however, the covid-affected cohort reported (marginally) lower well-being than the pre-covid cohort (H1d). Second, longitudinal analyses showed that the well-being and self-efficacy of covid-affected students decreased throughout the first semester and increased again in the second semester (H2a–H2b).

Next to the hypotheses testing, this study also offers relevant insights irrespective of COVID-19. The correlation coefficients demonstrate that the central

variables are closely interrelated. When arriving at university, students' well-being is positively correlated to their academic confidence and feeling of informedness. This implies that students with higher well-being tend to also feel more confident and informed when starting their university journey. Furthermore, self-efficacy is significantly correlated with well-being and academic confidence, as expected based on Gutiérrez and Tomás (2019) and St Clair-Thompson et al. (2017). Hence, when universities aim to enhance students' transition to higher education, all these variables should be considered. Additionally, the correlations offer insight into a gender gap: female students tend to have lower well-being, academic confidence and self-efficacy than male students, consistent with previous research (Lau et al., 2019; Michie et al., 2001; Savage et al., 2020). University educators should be aware of these potential gender differences.

The Effect of COVID-19 on Students' Arrival at University

Contrary to expectations (H1a–H1c), cross-sectional data analyses do not reveal significant differences between the covid-affected and pre-covid cohort, regarding students' initial experiences on arrival at university. Similar levels of well-being, academic confidence, and feeling of informedness imply that covid-affected students were not disadvantaged in their initial university transition. Other scholarly views on this are, however, ambiguous. Some scholars described that derangements in well-being are exacerbated at the start of the new academic year after a less restful vacation period, given the many canceled plans during the summer break of 2020 (Tasso et al., 2021). However, this was not the case for this study's covid-affected students. Contrarily, this study provides evidence on students' mental recovery during summer break, after a tumultuous completion of their secondary education. Despite additional social and preparatory challenges characterizing their transition during the pandemic, students

seem to have effective coping strategies to overcome these difficulties (Laher et al., 2021). These empirical results complement those of Schwartz et al. (2021), who found acceptable levels of students' mental health during their return to school in September 2020. These authors demonstrated that, six months after the pandemic started, students generally recovered sufficiently from, or at least coped with, post-traumatic stress symptoms. Moreover, Charles et al. (2021) have argued that, by fall 2020, the prevalence of mental health problems among university students returned to pre-pandemic levels, despite the pandemic's disruptive effect on their well-being in spring 2020. These researchers claimed that, after six months, students were coping well with the pandemic, resulting in largely resolved psychological impairments despite ongoing public health measures. Additionally, Liste et al. (2021) demonstrated that the personal life satisfaction of undergraduate students recovered, after a dramatic decline in spring 2020. Hence, literature pointed out students' resilience in dealing with the long-term COVID-19 implications. The present study adds to this, by focusing on transitioning first-year students and by comparing two cohorts cross-sectionally: a covid-affected and a pre-covid cohort.

Since transitioning to university should not be solely reduced to the moment of arrival at university (Brooman & Darwent, 2014), this study also aims to identify differences in well-being between the pre-covid and covid-affected cohort in the third week of the academic year. Analyses showed that students from both cohorts experienced decreased well-being after three weeks, compared to their arrival at university. This decrease during the initial weeks has been explained by first-year students' greater responsibilities and misguided expectations of the transition (Conley et al., 2014; Cooke et al., 2006). Regarding H1d, a marginal difference in well-being was observed in the third week of the first semester: pandemic-related restrictions and the

resulting duration effect (de Man et al., 2021) appear to have widened the gap between the covid-affected and pre-covid cohorts. Consequently, the observed decline in well-being after three weeks was more problematic for the covid-affected cohort. Another explanation could be that the measures against COVID-19 became more stringent after three weeks (Appendix A), which could have put additional pressure on students. In sum, based on the cross-sectional analyses, universities are advised to extend the induction period for their first-year students, and to provide additional support during their first weeks (e.g., social events) to improve student well-being, also in post-pandemic times.

Well-Being and Self-Efficacy During a Covid-Affected Academic Year

Concerning the longitudinal analyses, this study provides several interesting findings. First, it sheds light on the relation between well-being and self-efficacy, given that both fluctuate very similarly throughout the academic year. The link between well-being and self-efficacy mirrors earlier findings on undergraduate students (Lau et al., 2019; Morton et al., 2014), specifically those from COVID-19 contexts (Sánchez-Teruel et al., 2021; Wen et al., 2021).

Second, the data demonstrated a decrease in well-being and self-efficacy during the first semester of a pandemic-affected academic year (H2a–H2b). This result corroborates the duration effect of pandemic-related restrictions, several months after the initial COVID-19 outbreak: the longer public health measures persist, the greater the negative effects on students (de Man et al., 2021; Savage et al., 2020). The trend of decreasing well-being and self-efficacy of first-year students calls for supplementary orientation initiatives and support incentives during the first semester.

The data, however, reveal a remarkable increase in well-being and self-efficacy from the end of the first semester to the start of the second semester. There are several

potential explanations for this. The results indicate that changes in well-being and self-efficacy—both negative and positive—coincide with the varying stringency of pandemic-related restrictions. Appendix A shows when the covid-affected cohort experienced tightening and loosening pandemic-related restrictions. Remarkably, when well-being and self-efficacy again increased at the start of the second semester, governmental restrictions against COVID-19 had been loosened shortly before. This prospect may have increased students' self-efficacy, allowing their well-being to rebound, after a first semester that was severely affected by the pandemic (Sánchez-Teruel et al., 2021; Wen et al., 2021). This study also demonstrates a similar mental recovery before, during summer break, as cross-sectional analyses revealed no significant differences between the pre-covid and covid-affected cohort regarding their well-being after summer break (H1a). Hence, after the first semester, the spring break in February 2021 could have enabled students to recover their well-being at the start of the second semester.

Limitations and Future Directions

This study has several limitations. First, the observed changes in well-being and self-efficacy during a covid-affected academic year could be explained by the rigor of pandemic containment measures. Nevertheless, the analyses did not include the perceived severeness of the measures. Moreover, the study design does not allow to investigate potential changes in well-being and self-efficacy throughout an academic year without COVID-19, as the incomplete split-plot factorial design only comprises two out of four measurement moments in the pre-covid period (Appendix B). Consequently, the observed changes during the covid-affected period cannot be cross-sectionally compared with complete pre-covid data. Irrespective of the pandemic, previous literature has suggested a decline in first-year students' well-being during the

first semester, followed by a stabilization or slight recovery in the second semester (e.g., Conley et al., 2014). As the process of university transition and adaptation entails a stressful period for many students, these fluctuations denote the uncertainty of new first-year experiences. This study indicates that the well-being of covid-affected students, although marginally, declined more drastically in the first three weeks of university, compared with pre-covid students. Accordingly, even given a similar potential change without the pandemic, one could speculate that COVID-19 amplified the negative trends. To accurately validate the variance that can be attributed to COVID-19, and to confirm if the pandemic amplified the stressors that generally characterize how students perceive their first year of undergoing study and assessment at higher education, future post-pandemic studies should further assess the changes in well-being and self-efficacy throughout the entire first year at university. Moreover, future research opportunities lie within the role of first-semester performance (Morton et al., 2014), and protective factors such as resilience (Laher et al., 2021) and coping strategies (Parada & Verlhias, 2022). These additional factors might contribute to the increased well-being at the start of the second semester, as observed in this study.

A second limitation of this study entails the operationalization of academic confidence and feeling of informedness, which were assessed by the single-item measures of Stankov et al. (2014) and Mouton and Ertl (2021), respectively. Given the comprehensive psychological notion of both, and the considerable correlations with well-being, these single-item measures might not have sufficient strength to fully capture these concepts. Hence, these variables are worth exploring further in university transition contexts.

Third, since this study focuses on first-year business economics students at one Western European university, the results might not be generalizable to all first-year

students. Future research could study the well-being of first-year students in other universities and disciplines.

Fourth, the longitudinal design and the survey's voluntary nature also add to the selectivity of the covid-affected sample: due to attrition, only 98 students reported data at all four measurement moments in the covid-affected period. To address this sample size limitation, the expectation maximization algorithm was used to estimate missing values. The longitudinal analyses give rise to future research questions. For instance, is the observed decrease in well-being and self-efficacy throughout students' first semester normative and inevitable, or do these non-cognitive traits stay stable for some students but not for others? If the latter is true, a profile study might explain individual differences, and what exactly did or did not contribute to this bounce-back of well-being and self-efficacy.

Finally, self-reported quantitative instruments could entail suboptimal detection of pandemic-induced psychological impairments (Laher et al., 2021). As the methods of this study allow to assess students' positive well-being, no conclusions can be drawn on the prevalence of particular mental health symptoms (e.g., anxiety, depression). Furthermore, combining quantitative with qualitative methods may be appropriate to capture the 'lived experience' of first-year students' well-being and self-efficacy, in response to COVID-19 and generally in the future.

Educational Implications

This study adds to the literature that has already signaled the problematic effect of COVID-19 on student well-being. The findings have several theoretical and practical implications for university educators and higher education authorities seeking to understand the experiences of first-year students, in the year following the initial COVID-19 outbreak. The long-term effects—rather than the instantaneous

consequences of the very first lockdown—combined with longitudinal methods (suggested by, for example, Sánchez-Teruel et al., 2021), represent a premise not tested previously. The need to persistently monitor students who were affected by the pandemic is emphasized by Pownall et al. (2022). These scholars stated that compromised university transition experiences could have a lasting disadvantage on students' entire academic trajectories. University educators and educational policies should acknowledge the need for additional support systems (e.g., structural policies to promote supportive student-staff and student-student interactions, interventions to strengthen students' coping skills), even after COVID-19 measures are revoked.

This study's insights have considerable added value, irrespective of COVID-19. First, as universities aim at improving students' transition to higher education, well-being, self-efficacy, academic confidence and feeling of informedness must be considered as important and interrelated concepts.

Second, there are inferences on students' e-learning experiences. Online and blended learning have been gaining ground in higher education for several years. However, according to the longitudinal analyses, well-being and self-efficacy are better when frequent on-campus learning and social contact are possible. This finding reinforces previous evidence (e.g., Liste et al., 2021): remote learning during the pandemic has been associated with poorer well-being and lower satisfaction of students. The discomfort of e-learning and subsequent social disengagement (Juntunen et al., 2022; Laher et al., 2021) may encourage universities to provide on-campus learning opportunities, and to refrain from adopting distance learning as the baseline teaching approach. Higher education authorities should therefore consider the important social benefits of on-campus learning in the ongoing debate on online learning.

Third, declining student well-being is not unique to the pandemic. Therefore, it is important that neither academic nor policy attention to well-being fades, once the pandemic has passed. COVID-19 merely exacerbated and rendered visible a problematic pattern that was already present in student populations (Conley et al., 2014). The repercussions of the pandemic on the well-being of first-year students have exacerbated the already vulnerable position their university transition entails. This study therefore identifies the need to support student well-being, also post-pandemic. Recommended practices from previous research include teaching approaches and safe learning environments that foster social interactions, as well as better access to support services such as counseling (e.g., Baik et al., 2019). Higher education authorities could consider different ways of supporting first-year students during their complex transition period, to improve their social, academic, and personal-emotional adaptation (St Clair-Thompson et al., 2017). To enhance students' social adaptation and sense of belonging, engaging orientation days or peer mentoring programs could be recommended. Furthermore, university authorities and their educational policies should also aid the academic adaptation of first-year students, by means of, for instance, study strategy workshops organized by specialized student counseling services. Counseling services (e.g., individual counseling sessions, support groups) are also a valuable institutional resource to structurally facilitate the personal-emotional adaptation of students. By offering various support initiatives (social, academic, etc.), universities should offer all students the choice to engage with those activities they might need or benefit most from.

This study demonstrates that university authorities should organize activities to increase first-year students' well-being throughout the entire first semester—rather than focusing solely on the initial moment of enrollment—since well-being and self-efficacy decline throughout the first semester. Structural support initiatives (such as peer

mentoring programs) could be implemented on a university-wide or department level, to mitigate this most vulnerable period before the first examination period in higher education. Moreover, universities should proactively stimulate students' protective resources, including self-efficacy, coping strategies, and resilience (Sánchez-Teruel et al., 2021), to address potential mental repercussions of the stressful transition to higher education. Targeted interventions (e.g., constructive feedback after assessments, resilience training) should encourage first-year students to adopt an academic mindset that is growth-oriented—as opposed to a fixed-trait mindset—enabling them to deal with study-related challenges and to overcome adversity, which, in turn, increases proactive coping and well-being (Juntunen et al., 2022; Parada & Verlhac, 2022).

The focus on student well-being, fueled by the COVID-19 crisis, should not wane in upcoming years. The tagline “never waste a good crisis” applies here, as universities are invited to rethink—and revise—their policies based on what the pandemic exposed as a key issue: the need to monitor and support students' well-being during their transition to higher education.

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Appendix A

Education-Specific Measures Experienced by the Covid-Affected Student Cohort (March 2020 – February 2021)

code green	Usual school year (2019–2020) before the outbreak of COVID-19.
code red	March 16, 2020: Start of national lockdown. Schools are closed; all classes are suspended. April 20, 2020: Secondary schools implement full-time online learning.
code orange	May 15, 2020: High school seniors are allowed to go to school for two days a week. The other part of classes is still organized online.
code yellow	July 1, 2020: Start of summer break. National health restrictions are less severe: e.g., traveling abroad is allowed, personal contact is allowed with up to 15 people, festivities such as wedding parties are allowed with 50 people.
code orange	September 21, 2020 (Semester ₁ Week ₁): Start of academic year 2020–2021. Switch to code orange in higher education. Lectures and practice sessions are allowed on campus with a capacity of 20%; facemasks are obligatory at all times. Public places such as libraries are open.
code red	October 26, 2020 (Semester ₁ Week ₆): Switch to <i>code red</i> in higher education. Lectures and practice sessions are no longer allowed on campus, but are to be organized online. Public spaces such as libraries are closed. More severe governmental restrictions are also imposed: restaurants and bars are closed, youth activities and organized sports activities are no longer allowed.
code red	January, 2021: Examination period. Some of the exams are organized online, some are is organized on campus under severe health restrictions.
code orange	February 8, 2021 (Semester ₂ Week ₁): Start of the second semester. Switch to <i>code orange</i> in higher education. Lectures are still to be organized online. Practice sessions and tutorials are allowed on campus, libraries and other public university locations are open.

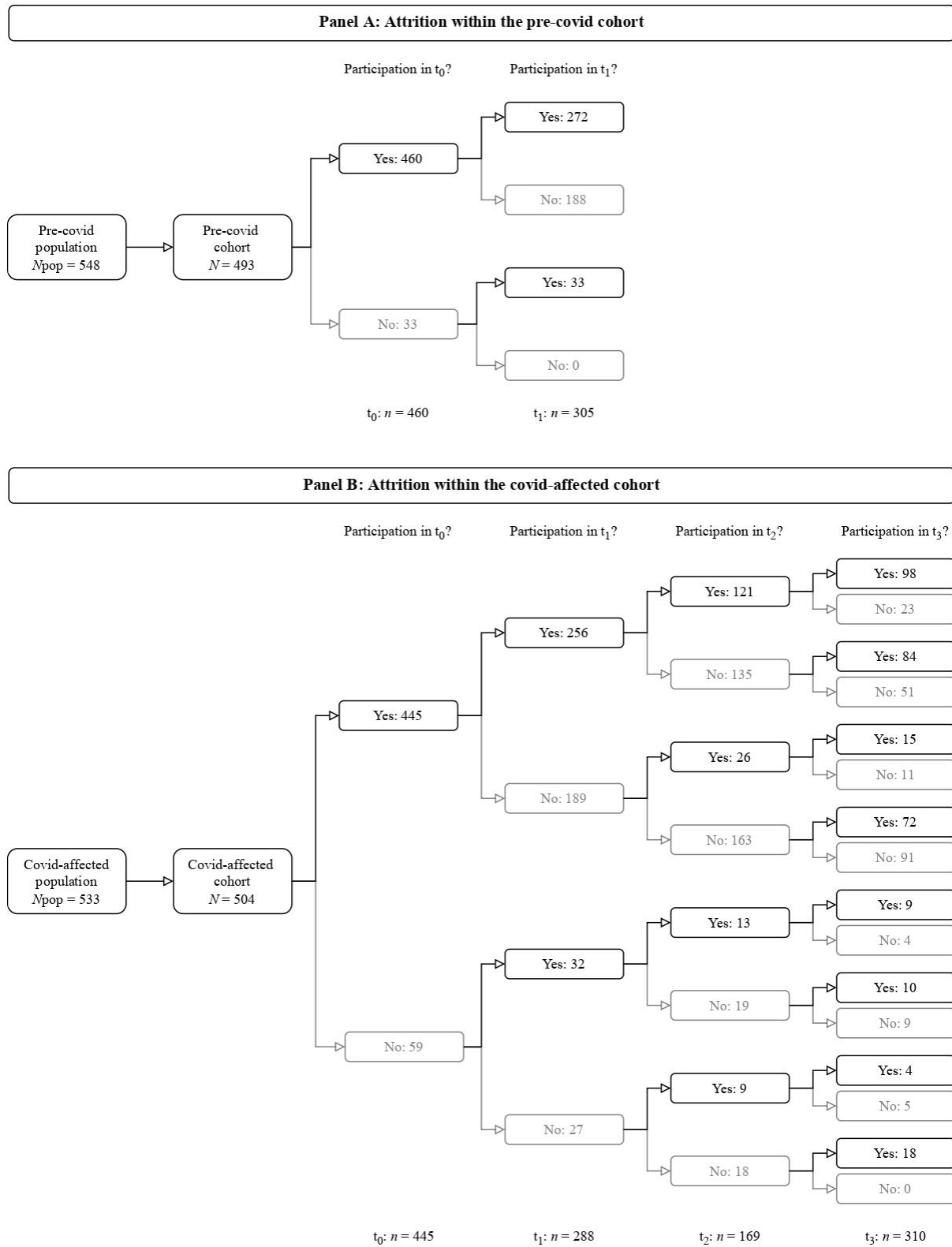
Appendix B*Incomplete Split-Plot 2×4 Factorial Design*

Measurement moment (between-subjects) Student cohort (within-subjects)	t ₀	t ₁	t ₂	t ₃
Pre-covid cohort	n = 460	n = 305		
Covid-affected cohort	n = 445	n = 288	n = 169	n = 310

Note: t₀ = Semester₁Week₀, t₁ = Semester₁Week₃, t₂ = Semester₁Week₁₁, t₃ = Semester₂Week₃

Appendix C

Sample Attrition



Note. t_0 = Semester₁Week₀, t_1 = Semester₁Week₃, t_2 = Semester₁Week₁₁, t_3 = Semester₂Week₃.

Appendix D

Survey Items

Variable	Item(s)	Measurement moment					
		Pre-covid		Covid-affected			
		t ₀	t ₁	t ₀	t ₁	t ₂	t ₃
Well-being	<i>I feel optimistic about the future.</i> <i>I feel relaxed.</i> <i>I feel interested in other people.</i> <i>I feel interested in new things.</i> <i>I feel confident.</i> <i>I feel good about myself.</i> On a 5-point Likert scale ^a Source: Tennant et al. (2007)	x	x	x	x	x	x
		x	x	x	x	x	x
		x	x	x	x	x	x
		x	x	x	x	x	x
		x	x	x	x	x	x
		x	x	x	x	x	x
		x	x	x	x	x	x
Academic confidence	How high do you estimate your own chances of success? 0-30%; 31-50%; 51-70%; 71-85%; 86-100%	x		x			
Feeling of informedness	I feel sufficiently informed to start this study program. On a 5-point Likert scale ^a	x		x			
Self-efficacy	<i>I can always manage to solve difficult problems if I try hard enough.</i> <i>If someone opposes me, I can find the means and ways to get what I want.</i> <i>It is easy for me to stick to my aims and accomplish my goals.</i> <i>I am confident that I could deal efficiently with unexpected events.</i> <i>Thanks to my resourcefulness, I know how to handle unforeseen situations.</i> <i>I can solve most problems if I invest the necessary effort.</i> <i>I can remain calm when facing difficulties because I can rely on my coping abilities.</i> <i>When I am confronted with a problem, I can usually find several solutions.</i> <i>If I am in trouble, I can usually think of a solution.</i> <i>I can usually handle whatever comes my way.</i> On a 4-point Likert scale ^b Source: Schwarzer & Jeruzalem (1995)			x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	
				x	x	x	

Note. t₀ = Semester₁Week₀, t₁ = Semester₁Week₃, t₂ = Semester₁Week₁₁, t₃ = Semester₂Week₃.

^a Verbal anchors of these items include the following: completely disagree (1); rather disagree (2); neutral (3); rather agree (4); completely agree (5).

^b Verbal anchors of these items include the following: not at all true (1); hardly true (2); moderately true (3); exactly true (4).